

Shreya Lanjewar

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EDUCATION

University of Pennsylvania, Philadelphia

Aug 2023 – May 2025

Master of Science in Mechanical Engineering and Applied Mechanics - Robotics concentration

Indian Institute of Technology - Mandi, India

Aug 2017 – May 2021

Bachelor of Technology in Mechanical Engineering

TECHNICAL SKILLS

Programming Languages: Python, Matlab, C/ C++, HTML

Frameworks and Libraries: NumPy, PyTorch, Pandas, openCV

Simulation and Robotics: ROS2, Webots, Matplotlib, SLAM, Drake, Gazebo

Micro-controllers: Arduino, ESP32, Jetson, Raspberry Pi3

Designing and Prototyping: SolidWorks, 3D printing, Laser Cutting, CAD, Catia, Ansys Workbench

EXPERIENCE

Research Assistant, ModLab, University of Pennsylvania, US

May 2024 - July 2025

- Developed algorithm for the vertical locomotion of a variable geometry truss in a 3D environment. Using kinematic and static contact analysis to design and configure the truss for optimal performance.

Teaching Assistant, Design of Mechatronics Systems, University of Pennsylvania, US

Aug - Dec 2024

- Provide guidance on interfacing analog components, microprocessor technology, and structured embedded programming, including register level programming.

PROJECTS

F1Tenth Autonomous Racing | ROS2 | TensorRT | SLAM |

Jan 2025 - Mar 2025

- Developed the control stack for autonomous F1 racing car of 1/10th scale with sensor fusion with Realsense camera and Hokuyo 2D Lidar.
- Implemented autonomous emergency braking (AEB), for safety and collision avoidance, path planning using RRT* and pure pursuit, SLAM for navigation

Trajectory Optimization and Path Planning for Underactuated Brachiation | Python | Docker | Drake | MPC |

Aug - Dec 2024

- Developed an optimized control-stack for 2-link brachiating robot traversing a 2D environment modeling energy-efficient swings.
- Implemented a hierarchical control structure with two levels. Low-level trajectory planner using direct collocation for hold-to-hold trajectories. High-level path planner using A* algorithm with energy-based edge costs.

Autonomous Navigation for Unstructured Environments | Python | pandas | openCV | Jetson |

Feb - May 2024

- Developed a vision-based autonomous navigation system using deep learning to assess terrain and plan paths.
- Implemented TRAVNET with ResNet18 for terrain mapping, generated safe and risky trajectory options, and deployed the system on a Jetson Xavier NX with an F1Tenth car, creating a relevant dataset for unstructured outdoor environments.

BUDD - Navigating in Indoor Environment | Python | ROS2 |

Oct - Feb 2024

- Developed BUDD's software architecture using ROS for efficient node communication and data integration. Implemented SLAM with LiDAR point clouds and a particle filter for precise localization and dynamic map updating.
- Designed a comprehensive navigation stack integrating localization nodes, cost map construction for obstacle avoidance, and sampling-based planners (RRT) with controllers for smooth, autonomous navigation to targets.

Pick and Place objects using 7-dof Franka Panda Arm | Python | ROS1 | openCV |

Oct - Dec 2023

- Implemented inverse kinematics for controlling a Franka Panda robotic arm to perform pick-and-place operations with two block types.
- Optimized motion trajectories for efficient task execution, achieving precise block stacking of four in approximately one minute.

Autonomous robot for Grand Theft Auto based competition | C | HTML | UDP | ESP32 |

Oct - Dec 2023

- Developed an autonomous robot with tank steering, custom gripper, and PID control for wall-following and beacon tracking. Implemented real-time control via a webpage and communication using UDP and ESPNOW.
- Optimized electronics systems with L298N motor drivers and sensors. Overcame design challenges with adaptive solutions, ensuring precise task execution.

Implementation of Neural Radiance Field | Python | PyTorch |

Nov - Dec 2023

- Implemented Neural Radiance Fields (NeRF) for 3D scene representation, fine-tuning parameters to achieve realistic rendering.
- Applied machine perception and computer vision techniques to enhance 3D scene accuracy and visual fidelity.

Two-view Stereo Algorithm | Python |

Nov - Dec 2023

- Developed a two-view stereo algorithm to reconstruct 3D scenes from calibrated stereo images, utilizing the Templering dataset for evaluation.
- Implemented image rectification, patch-based disparity computation, and depth map generation using matching metrics (SSD, SAD, and ZNCC) to accurately capture 3D structures from 2D viewpoints.